

Lösungen: Übungsblatt Grundlagen

14.10.2026

Aufgabe 1)

- a) $3(x + 4) = 3x + 12$
- b) $-5(2y - 7) = -10y + 35$
- c) $2a(3a - 4b + 2) = 6a^2 - 8ab + 4a$
- d) $-(x - 2y + 5) = -x + 2y - 5$

Aufgabe 2)

- a) $15x + 20 = 5(3x + 4)$
- b) $12a^2 - 18a = 6a(2a - 3)$
- c) $14xy - 21x^2 + 7x = 7x(2y - 3x + 1)$
- d) $24mn^2 + 16m^2n = 8mn(3n + 2m)$

Aufgabe 3)

- a)
$$\begin{aligned} 4(2x - 3) - 2(3x - 5) &= 8x - 12 - (6x - 10) \\ &= 8x - 12 - 6x + 10 \\ &= 2x - 2 \end{aligned}$$
- b)
$$\begin{aligned} 3x(x - 2) - x(2x + 1) + 5x &= 3x^2 - 6x - (2x^2 + x) + 5x \\ &= 3x^2 - 6x - 2x^2 - x + 5x \\ &= x^2 - 2x \end{aligned}$$